MISALIGNMENT SENSITIVITY THRESHOLDS FOR SOFTMAX-PARAMETERIZED TRADING AGENTS

NEURICO

Abstract. We study the problem of quantifying the vulnerability of softmax-parameterized trading agents to misalignment, defined as behavioral deviation from intended objectives induced by adversarial reward perturbations. We introduce the *misalignment sensitivity threshold* $\tau(\delta)$, the minimum perturbation intensity required to produce a KL divergence exceeding δ between the perturbed and baseline policies. Our main result establishes that $\tau(\delta) = \sqrt{2\delta/\|G\|_F^2} + O(\delta)$, where $\|G\|_F^2$ is the Fisher information norm of the policy gradient sensitivity. We prove that the Fisher information norm scales as $\|G\|_F^2 \propto 1/T^2$, yielding the temperature scaling law $\tau \propto T$. This identifies the softmax temperature as the primary determinant of misalignment robustness, with lower temperatures producing greater vulnerability. We further show that market volatility σ and decision horizon h have asymptotically negligible effects on the threshold, entering only at order $O(\sigma^{-2})$ and $O(h^{-1})$ respectively. These theoretical predictions are verified computationally across more than one hundred MDP configurations, confirming the linear temperature scaling ($R^2 = 0.999$) and the negligible dependence on market parameters ($|\alpha|, |\beta| < 0.02$). Our results redirect the focus of misalignment safety from external market conditions to internal policy architecture, providing an actionable metric for deployment decisions.

1. Introduction

Large language model (LLM) based agents are increasingly deployed as autonomous decision-makers in financial markets, where they execute trading strategies with minimal human oversight [6, 2]. In such settings, the agent optimizes a proxy reward that approximates, but does not perfectly capture, the principal's true objective. When the proxy reward is susceptible to perturbation—whether through adversarial manipulation, distributional shift, or specification error—the agent's behavior may deviate from the intended objective, a phenomenon broadly termed *misalignment*. Given the economic stakes of automated trading, a quantitative understanding of when misalignment becomes behaviorally significant is essential for responsible deployment.

The theoretical foundations of misalignment have advanced considerably. Skalse et al. [4] proved that reward hacking is generically unavoidable: over any open set of stochastic policies, every non-trivial proxy reward is hackable. Wang and Huang [5] showed that distortion is inevitable under finite evaluation, and conjectured the existence of a phase transition between gaming and degrading regimes. On the empirical side, Debenedetti et al. [1] established a power-law relationship between

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compute and adversarial robustness with a remarkably small exponent ($\alpha \approx 0.01$), while Nathanson et al. [3] demonstrated logarithmic scaling of adversarial harm with model size ratios.

Despite these advances, no existing framework provides a quantitative answer to the following question: *given a trading agent with specific policy parameters operating in a market with given volatility and decision horizon, what is the minimum perturbation intensity that induces statistically significant misalignment?* The impossibility results of [4] guarantee that misalignment can occur but do not predict the perturbation magnitude required. The distortion analysis of [5] characterizes the direction of misalignment but not the threshold intensity. The empirical scaling laws of [1, 3] lack theoretical derivation and do not connect to market-specific parameters.

In this paper, we develop a rigorous mathematical framework for *misalignment sensitivity thresholds* in the context of softmax-parameterized trading agents operating in finite Markov decision processes (MDPs). Our contributions are as follows.

- We introduce the misalignment sensitivity threshold $\tau(\delta)$, defined as the infimum perturbation intensity ε for which the KL divergence between perturbed and baseline policies exceeds a significance level δ (Definition 2.7).
- We prove that $\tau(\delta)$ exists and is finite for any finite MDP with softmax policies (Theorem 3.1), and characterize it via the Fisher information of the policy parameterization: $\tau(\delta) = \sqrt{2\delta/\|G\|_F^2} + O(\delta)$ (Theorem 3.4).
- We establish the temperature scaling law $\tau \propto T$ (Corollary 3.6), identifying the softmax temperature as the dominant determinant of misalignment robustness.
- We prove that market volatility σ and decision horizon h are asymptotically negligible, with $\tau(\sigma) = \tau_0 + O(\sigma^{-2})$ and $\tau(h) = \tau_0 + O(h^{-1})$ (Propositions 3.8 and 3.9), refuting the initially hypothesized power-law dependence $\tau \propto \sigma^{-\alpha} h^{-\beta}$.
- We verify all theoretical predictions computationally across more than one hundred MDP configurations, confirming the linear temperature scaling with $R^2 = 0.999$ and the negligible market-parameter exponents $|\alpha|, |\beta| < 0.02$.

The proofs combine perturbation analysis of softmax policies, second-order Taylor expansion of KL divergence, and the standard softmax gradient identity. The key technical insight is that the Fisher information of a softmax distribution concentrates the sensitivity of the policy in a single parameter—the temperature T—that governs the "peakedness" of the decision distribution. Market parameters affect the Q-values but not the mechanism by which perturbations propagate through the softmax, leading to their asymptotic irrelevance.

The paper is organized as follows. Section 2 establishes notation and defines the trading MDP, intervention operators, and the misalignment threshold. Section 3 states and proves the main results. Section 4 discusses implications, connections to existing work, and open questions. Section 5 summarizes our contributions.

2. Preliminaries

We work throughout with finite MDPs. All logarithms are natural. For a finite set X, we write $\Delta(X)$ for the set of probability distributions on X. For a distribution $p \in \Delta(X)$, we write $H(p) = -\sum_x p(x) \log p(x)$ for the Shannon entropy

and $D_{\mathrm{KL}}(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)}$ for the Kullback–Leibler divergence. We use the convention $0 \log 0 = 0$.

2.1. **Trading MDPs.**

Definition 2.1 (Trading MDP)**.** A *trading MDP* is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}_\sigma, R, \gamma, h)$ where:

(i) $\mathcal{S}$ is a finite state space;
(ii) $\mathcal{A}$ is a finite action space;
(iii) $\mathcal{T}_\sigma \colon \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0,1]$ is a volatility-parameterized transition function, with $\sigma > 0$;
(iv) $R \colon \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is a bounded reward function;
(v) $\gamma \in (0,1)$ is the discount factor;
(vi) $h = 1/(1-\gamma)$ is the effective decision horizon.

We impose a regularity condition on the transition function that formalizes the intuition that higher volatility produces less predictable transitions.

Definition 2.2 (Volatility-parameterized transitions)**.** The transition function $\mathcal{T}_\sigma$ is *volatility-parameterized* if:

(i) $\mathcal{T}_\sigma(s' \mid s, a)$ is continuous in σ for all $s, s' \in \mathcal{S}$ and $a \in \mathcal{A}$;
(ii) $H(\mathcal{T}_\sigma(\cdot \mid s, a))$ is non-decreasing in σ for all $s \in \mathcal{S}$ and $a \in \mathcal{A}$.

A canonical example is the discretized Gaussian kernel: for a one-dimensional state space $\mathcal{S} = \{s_1, \dots, s_n\}$ with mean transition target $\mu(s,a)$, define

$$\mathcal{T}_\sigma(s_j \mid s_i, a) = \frac{\exp\bigl(-(s_j - \mu(s_i,a))^2/(2\sigma^2)\bigr)}{\sum_{k=1}^{n} \exp\bigl(-(s_k - \mu(s_i,a))^2/(2\sigma^2)\bigr)}. \tag{2.1}$$

This satisfies Definition 2.2: as $\sigma \to \infty$, the kernel converges to the uniform distribution on $\mathcal{S}$.

2.2. **Interventions and softmax policies.**

Definition 2.3 (Intervention operator)**.** For intensity $\varepsilon \geq 0$ and a *perturbation direction* $\Phi \colon \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ with $\|\Phi\|_\infty = 1$, the *perturbed proxy reward* is

$$R_\varepsilon(s,a) = R_{\mathrm{proxy}}(s,a) + \varepsilon\, \Phi(s,a).$$

Definition 2.4 (Softmax optimal policy)**.** For temperature $T > 0$, the *softmax optimal policy* under reward R is

$$\pi_R^T(a \mid s) = \frac{\exp\bigl(Q_R(s,a)/T\bigr)}{\sum_{b \in \mathcal{A}} \exp\bigl(Q_R(s,b)/T\bigr)}, \tag{2.2}$$

where Q_R is the optimal Q-function under R, satisfying

$$Q_R(s,a) = R(s,a) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{T}_\sigma(s' \mid s, a) \max_{b \in \mathcal{A}} Q_R(s', b).$$

Remark 2.5. The softmax policy π_R^T depends on Q_R only through the differences $Q_R(s,a) - Q_R(s,b)$, since adding a state-dependent constant $c(s)$ to all Q-values leaves π_R^T unchanged. This shift-invariance is central to the weak dependence of the threshold on market parameters established in Section 3.

2.3. **Policy divergence and the misalignment threshold.** We write $\pi_\varepsilon = \pi^T_{R_\varepsilon}$ for the softmax policy under the perturbed reward R_ε, and $\pi_0 = \pi^T_{R_{\text{proxy}}}$ for the unperturbed baseline.

Definition 2.6 (Policy divergence)**.** The *policy divergence* at perturbation intensity ε is

$$D(\varepsilon) = \sum_{s\in\mathcal{S}} d^{\pi_\varepsilon}(s) \sum_{a\in\mathcal{A}} \pi_\varepsilon(a \mid s) \log \frac{\pi_\varepsilon(a \mid s)}{\pi_0(a \mid s)}, \tag{2.3}$$

where $d^\pi(s)$ is the stationary state distribution under policy π.

Definition 2.7 (Misalignment sensitivity threshold)**.** For significance level $\delta > 0$, the *misalignment sensitivity threshold* is

$$\tau(\delta) = \inf\bigl\{\varepsilon > 0 : D(\varepsilon) > \delta\bigr\}. \tag{2.4}$$

Informally, $\tau(\delta)$ measures the minimum adversarial perturbation required to produce a statistically significant change in agent behavior. Small τ indicates high vulnerability; large τ indicates robustness.

2.4. **Policy gradient sensitivity and Fisher information.** We define two quantities that play a central role in the characterization of τ.

Definition 2.8 (Policy gradient sensitivity)**.** The *policy gradient sensitivity* is

$$G(s,a) = \left.\frac{\partial \pi^T_{R_\varepsilon}(a \mid s)}{\partial \varepsilon}\right|_{\varepsilon=0}. \tag{2.5}$$

Definition 2.9 (Fisher information norm)**.** The *Fisher information norm* of G is

$$\|G\|_F^2 = \sum_{s\in\mathcal{S}} d^{\pi_0}(s) \sum_{a\in\mathcal{A}} \frac{G(s,a)^2}{\pi_0(a \mid s)}. \tag{2.6}$$

Remark 2.10. The quantity $\|G\|_F^2$ is the Fisher information of the one-parameter family $\{\pi_\varepsilon\}_{\varepsilon\geq 0}$ evaluated at $\varepsilon = 0$, averaged over the stationary state distribution. It measures the sensitivity of the policy to infinitesimal reward perturbations in the direction Φ.

3. Main results

3.1. **Existence and characterization of the threshold.** Our first result establishes that the misalignment sensitivity threshold is well-defined.

Theorem 3.1 (Existence)**.** *Let $\mathcal{M}$ be a finite trading MDP with softmax policy parameterization at temperature $T > 0$. For any perturbation direction Φ with $\|\Phi\|_\infty = 1$ and significance level $\delta > 0$, the misalignment sensitivity threshold $\tau(\delta)$ exists and satisfies $0 < \tau(\delta) < \infty$.*

Proof. We establish each bound separately.

Lower bound ($\tau > 0$). The Q-function $Q_{R_\varepsilon}(s,a)$ is the unique fixed point of the Bellman operator with reward $R_\varepsilon = R_{\text{proxy}} + \varepsilon\Phi$. Since Φ is bounded, the Bellman operator depends continuously on ε, and by the contraction mapping theorem the fixed point Q_{R_ε} is continuous in ε. The softmax map $Q \mapsto \pi^T$ is smooth for $T > 0$ (it is a composition of exponentials and division by a positive sum). Therefore $\pi_\varepsilon = \pi^T_{R_\varepsilon}$ is continuous in ε, and hence $D(\varepsilon) = D_{\text{KL}}(\pi_\varepsilon \| \pi_0)$ is continuous in ε

with $D(0) = 0 < \delta$. By continuity, there exists $\varepsilon_0 > 0$ such that $D(\varepsilon) < \delta$ for all $\varepsilon \in [0, \varepsilon_0)$, whence $\tau \geq \varepsilon_0 > 0$.

Upper bound ($\tau < \infty$). For ε sufficiently large, the term $\varepsilon\Phi$ dominates R_{proxy} in the Bellman equations, so the Q-function of R_ε is asymptotically determined by Φ alone. Concretely, for any state s where $\operatorname{argmax}_a \Phi(s,a)$ is a singleton not equal to $\operatorname{argmax}_a Q_{R_{\text{proxy}}}(s,a)$, the softmax policy $\pi_\varepsilon(\cdot \mid s)$ concentrates on $\operatorname{argmax}_a \Phi(s,a)$ as $\varepsilon \to \infty$, producing a limiting policy π_∞ with $D_{\text{KL}}(\pi_\infty \| \pi_0) > 0$. For generic Φ (which is dense in the unit ball of ℓ^∞), such states exist. Therefore there exists $\varepsilon_1 < \infty$ with $D(\varepsilon_1) > \delta$, giving $\tau \leq \varepsilon_1 < \infty$. □

The following lemma computes the first-order response of the softmax policy to reward perturbations.

Lemma 3.2 (Policy sensitivity formula)**.** *For a softmax policy π^T at temperature $T > 0$ with Q-function Q, the policy gradient sensitivity is*

$$G(s,a) = \frac{1}{T}\,\pi^T(a \mid s)\Big(\widetilde{\Phi}(s,a) - \mathbb{E}_{\pi^T(\cdot|s)}\big[\widetilde{\Phi}(s,\cdot)\big]\Big), \tag{3.1}$$

where $\widetilde{\Phi}(s,a) = \partial Q_{R_\varepsilon}(s,a)/\partial\varepsilon\big|_{\varepsilon=0}$ is the Q-function sensitivity to the perturbation.

Proof. The softmax policy is

$$\pi_\varepsilon^T(a \mid s) = \frac{\exp\big(Q_{R_\varepsilon}(s,a)/T\big)}{\sum_{b\in\mathcal{A}} \exp\big(Q_{R_\varepsilon}(s,b)/T\big)}.$$

This is the standard softmax function applied to the vector $Q_{R_\varepsilon}(s,\cdot)/T$. The derivative of the softmax output $z_i = e^{x_i}/\sum_j e^{x_j}$ with respect to a parameter θ is the classical identity

$$\frac{\partial z_i}{\partial\theta} = z_i\left(\frac{\partial x_i}{\partial\theta} - \sum_j z_j \frac{\partial x_j}{\partial\theta}\right). \tag{3.2}$$

Applying (3.2) with $z_i = \pi^T(a \mid s)$, $x_i = Q_{R_\varepsilon}(s,a)/T$, and $\theta = \varepsilon$, we obtain

$$G(s,a) = \pi^T(a \mid s)\left(\frac{1}{T}\widetilde{\Phi}(s,a) - \sum_b \pi^T(b \mid s)\frac{1}{T}\widetilde{\Phi}(s,b)\right),$$

which is (3.1). □

Remark 3.3. The Q-function sensitivity $\widetilde{\Phi}$ incorporates both the direct effect of the perturbation Φ at the current state and its discounted propagation through future states. In many formulas below, we write Φ in place of $\widetilde{\Phi}$ for notational clarity, with the understanding that all statements hold for the effective perturbation direction.

We now state and prove the main characterization theorem.

Theorem 3.4 (Threshold characterization)**.** *Under the conditions of Theorem 3.1, the misalignment sensitivity threshold satisfies*

$$\tau(\delta) = \sqrt{\frac{2\delta}{\|G\|_F^2}} + O(\delta) \tag{3.3}$$

as $\delta \to 0$, where $\|G\|_F^2$ is the Fisher information norm of the policy gradient sensitivity.

Proof. The proof proceeds in three steps: we expand $D(\varepsilon)$ to second order, identify the leading term with the Fisher information, and invert the relation.

Step 1: Taylor expansion of the per-state KL divergence. Fix a state $s \in \mathcal{S}$ and write $\pi_\varepsilon = \pi_\varepsilon(\cdot \mid s)$, $\pi_0 = \pi_0(\cdot \mid s)$, $G = G(s, \cdot)$ for brevity. The first-order expansion gives

$$\pi_\varepsilon(a \mid s) = \pi_0(a \mid s) + \varepsilon\, G(s, a) + O(\varepsilon^2).$$

The per-state KL divergence is

$$D_{\mathrm{KL}}(\pi_\varepsilon(\cdot \mid s) \| \pi_0(\cdot \mid s)) = \sum_a \pi_\varepsilon(a \mid s) \log \frac{\pi_\varepsilon(a \mid s)}{\pi_0(a \mid s)}.$$

Substituting $\pi_\varepsilon = \pi_0 + \varepsilon G + O(\varepsilon^2)$ and using $\log(1 + x) = x - x^2/2 + O(x^3)$ with $x = \varepsilon G(s, a)/\pi_0(a \mid s)$, we obtain

$$\begin{aligned} \sum_a \pi_\varepsilon(a \mid s) \log \frac{\pi_\varepsilon(a \mid s)}{\pi_0(a \mid s)} &= \sum_a \big(\pi_0 + \varepsilon G\big) \left(\frac{\varepsilon G}{\pi_0} - \frac{\varepsilon^2 G^2}{2\pi_0^2} + O(\varepsilon^3) \right) \\ &= \varepsilon \sum_a G(s, a) + \frac{\varepsilon^2}{2} \sum_a \frac{G(s, a)^2}{\pi_0(a \mid s)} + O(\varepsilon^3). \end{aligned} \tag{3.4}$$

Step 2: The first-order term vanishes. Since $\pi_\varepsilon(\cdot \mid s)$ is a probability distribution for every ε, differentiating $\sum_a \pi_\varepsilon(a \mid s) = 1$ gives $\sum_a G(s, a) = 0$. Therefore the first-order term in (3.4) vanishes, and

$$D_{\mathrm{KL}}(\pi_\varepsilon(\cdot \mid s) \| \pi_0(\cdot \mid s)) = \frac{\varepsilon^2}{2} \sum_a \frac{G(s, a)^2}{\pi_0(a \mid s)} + O(\varepsilon^3). \tag{3.5}$$

Step 3: Averaging and inversion. Averaging (3.5) over the stationary distribution and noting that $d^{\pi_\varepsilon}(s) = d^{\pi_0}(s) + O(\varepsilon)$, we obtain

$$D(\varepsilon) = \frac{\varepsilon^2}{2} \|G\|_F^2 + O(\varepsilon^3).$$

Setting $D(\varepsilon) = \delta$ and solving for ε, we find $\varepsilon^2 = 2\delta/\|G\|_F^2 + O(\delta^{3/2})$, whence

$$\tau(\delta) = \sqrt{\frac{2\delta}{\|G\|_F^2}} + O(\delta). \qquad \square$$

Example 3.5. Consider a two-state, two-action MDP with $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{a_1, a_2\}$, Q-values $Q(s_1, a_1) = 1$, $Q(s_1, a_2) = 0$, $Q(s_2, a_1) = 0$, $Q(s_2, a_2) = 1$, perturbation $\Phi(s, a) = Q(s, a)$, uniform stationary distribution $d(s) = 1/2$, and temperature T. Then

$$\pi^T(a_1 \mid s_1) = \frac{e^{1/T}}{e^{1/T} + 1}, \qquad G(s_1, a_1) = \frac{1}{T} \cdot \frac{e^{1/T}}{(e^{1/T} + 1)^2}.$$

The Fisher information norm is $\|G\|_F^2 = \frac{1}{T^2} \cdot \frac{e^{1/T}}{(e^{1/T}+1)^2}$. For $T = 1$, this gives $\|G\|_F^2 \approx 0.197$ and $\tau(0.01) \approx \sqrt{0.02/0.197} \approx 0.319$.

3.2. **Temperature scaling.**

Corollary 3.6 (Temperature scaling law)**.** *The Fisher information norm scales as* $\|G\|_F^2 \propto 1/T^2$*, and consequently*

$$\tau(\delta) \propto T. \tag{3.6}$$

Proof. From Lemma 3.2,

$$G(s,a) = \frac{1}{T}\,\pi^T(a \mid s)\big(\Phi(s,a) - \mathbb{E}_{\pi^T}[\Phi(s,\cdot)]\big).$$

Substituting into the Fisher information norm (2.6):

$$\begin{aligned}
\|G\|_F^2 &= \sum_s d(s) \sum_a \frac{1}{\pi_0(a \mid s)} \cdot \frac{1}{T^2}\pi_0(a \mid s)^2\big(\Phi(s,a) - \mathbb{E}_{\pi_0}[\Phi]\big)^2 \\
&= \frac{1}{T^2}\sum_s d(s) \sum_a \pi_0(a \mid s)\big(\Phi(s,a) - \mathbb{E}_{\pi_0}[\Phi]\big)^2 \\
&= \frac{1}{T^2}\sum_s d(s)\ \mathrm{Var}_{\pi_0(\cdot|s)}\big(\Phi(s,\cdot)\big). \qquad (3.7)
\end{aligned}$$

The variance term $V := \sum_s d(s)\ \mathrm{Var}_{\pi_0(\cdot|s)}(\Phi(s,\cdot))$ depends on T through the policy π_0, but the dominant scaling is captured by the explicit $1/T^2$ prefactor. We verify this in two regimes.

High temperature ($T \gg \Delta Q$, where $\Delta Q = \max_{s,a} Q(s,a) - \min_{s,a} Q(s,a)$): the policy approaches uniform, so $V \to \mathrm{Var}_{\mathrm{unif}}(\Phi)$, which is a positive constant independent of T. Thus $\|G\|_F^2 \sim V_\infty/T^2$.

Low temperature ($T \ll \Delta Q$): the policy concentrates on the greedy action, and $\mathrm{Var}_{\pi_0}(\Phi)$ depends on how Φ values compare at the greedy action versus others. For generic Φ, V remains bounded and bounded away from zero. The leading-order behavior $\|G\|_F^2 \sim C/T^2$ with $C = C(Q,\Phi) > 0$ persists.

In both regimes, $\|G\|_F^2 = C(Q,\Phi)/T^2$ at leading order. By Theorem 3.4,

$$\tau(\delta) = \sqrt{\frac{2\delta}{\|G\|_F^2}} = T\sqrt{\frac{2\delta}{C(Q,\Phi)}} \propto T. \qquad \square$$

Remark 3.7. Corollary 3.6 reveals a fundamental *robustness–capability tradeoff*: increasing T raises the threshold τ (more robust to misalignment) but also increases the entropy of the policy (less decisive, lower expected reward). This tradeoff is intrinsic to softmax parameterization and cannot be circumvented within this class of policies.

3.3. **Weak dependence on market parameters.**

We now show that the market parameters σ (volatility) and h (decision horizon) enter the threshold only at subleading order.

Proposition 3.8 (Weak volatility dependence)**.** *For a trading MDP with Gaussian transition kernel* (2.1) *of width* σ*, the threshold satisfies*

$$\tau(\sigma) = \tau_0 + O(\sigma^{-2}) \tag{3.8}$$

as $\sigma \to \infty$*, where* τ_0 *is the threshold in the uniform-transition limit.*

Proof. The dependence of $\|G\|_F^2$ on σ enters through the Q-values $Q_R(s,a)$, which are determined by the Bellman equation involving $\mathcal{T}_\sigma$, and through the stationary distribution d^{π_0}.

For the Gaussian kernel (2.1), as $\sigma \to \infty$ the transition probabilities converge to the uniform distribution: $\mathcal{T}_\sigma(s' \mid s,a) \to 1/|\mathcal{S}|$. The rate of convergence is determined by the Gaussian tail: for fixed s, s', a,

$$\mathcal{T}_\sigma(s' \mid s,a) = \frac{1}{|\mathcal{S}|} + O(\sigma^{-2})$$

since the exponential weights $\exp(-(s_j-\mu)^2/(2\sigma^2))$ expand as $1-(s_j-\mu)^2/(2\sigma^2)+O(\sigma^{-4})$.

The Bellman operator is a contraction with modulus $\gamma < 1$, and the Q-function depends Lipschitz-continuously on the transition kernel with constant $R_{\max}/(1-\gamma)^2$, where $R_{\max} = \max_{s,a}|R(s,a)|$. Therefore

$$Q_\sigma(s,a) - Q_\infty(s,a) = O(\sigma^{-2}).$$

Since the Fisher information norm $\|G\|_F^2$ depends on Q_σ through the softmax policy, which is a smooth function of the Q-values, we conclude

$$\|G\|_F^2(\sigma) = \|G\|_F^2(\infty) + O(\sigma^{-2}),$$

and therefore $\tau(\sigma) = \tau_\infty + O(\sigma^{-2})$. □

Proposition 3.9 (Weak horizon dependence)**.** *For effective horizon $h = 1/(1-\gamma)$, the threshold satisfies*

$$\tau(h) = \tau_0 + O(h^{-1}) \tag{3.9}$$

as $h \to \infty$.

Proof. The discount factor is $\gamma = 1 - 1/h$. The Q-function satisfies

$$Q(s,a) = R(s,a) + \gamma \sum_{s'} \mathcal{T}_\sigma(s' \mid s,a) V(s'),$$

where $V(s) = \max_a Q(s,a)$. As $\gamma \to 1$ (equivalently $h \to \infty$), the Q-values grow as $Q \sim R_{\text{avg}} \cdot h$, but the Q-value *differences* $Q(s,a) - Q(s,b)$—which are the only quantities entering the softmax policy (Remark 2.5)—converge to a finite limit. Indeed,

$$Q(s,a) - Q(s,b) = R(s,a) - R(s,b) + \gamma \sum_{s'} \big[\mathcal{T}(s' \mid s,a) - \mathcal{T}(s' \mid s,b)\big] V(s'),$$

and the value differences $V(s') - V(s'')$ remain bounded by $R_{\max}/(1-\gamma)$ but cancel in the sum structure, yielding convergence of Q-differences at rate $O(\gamma) = O(1-1/h)$.

Since the softmax policy depends only on Q-differences (which converge as $h \to \infty$), the Fisher information norm converges:

$$\|G\|_F^2(h) = \|G\|_F^2(\infty) + O(h^{-1}),$$

and therefore $\tau(h) = \tau_\infty + O(h^{-1})$. □

Example 3.10. Consider the two-state MDP of Example 3.5 with a Gaussian transition kernel of width σ. For $T = 0.3$ and $\delta = 0.01$, the theoretical threshold is $\tau \approx 0.14$ for all $\sigma \in [0.05, 50]$, with a variation of less than 2% across three orders

of magnitude in σ. This illustrates the content of Proposition 3.8: the volatility parameter is essentially irrelevant to misalignment vulnerability.

3.4. **Computational verification.** We verify the theoretical predictions on finite trading MDPs with state space of size $n_{\text{prices}} \times n_{\text{positions}}$, action space $\mathcal{A} = \{\text{sell}, \text{hold}, \text{buy}\}$, Gaussian transition kernels, and softmax policies. The threshold is computed numerically via bisection over random perturbation directions. Theoretical predictions use $\tau_{\text{theory}} = \sqrt{2\delta/F_{\max}}$ where $F_{\max}$ is the Fisher information evaluated at the worst-case perturbation direction.

Temperature dependence. Table 1 reports thresholds for $T \in [0.05, 3.0]$. A power-law fit yields $\tau_{\text{numerical}} \propto T^{1.008}$ with $R^2 = 0.9992$ and $p < 10^{-13}$, confirming the linear scaling of Corollary 3.6. The theoretical prediction $\tau_{\text{theory}} \propto T^{0.987}$ has $R^2 = 0.9993$. Numerical and theoretical values agree within 10% across two orders of magnitude in T.

TABLE 1. Temperature dependence of the misalignment threshold. Power-law fit: $\tau \propto T^{1.008}$, $R^2 = 0.9992$.

T	τ (numerical)	τ (theoretical)	$\|G\|_F^2$
0.05	0.0219	0.0215	43.24
0.10	0.0431	0.0466	9.21
0.20	0.0803	0.0879	2.59
0.50	0.2148	0.2319	0.37
1.00	0.4260	0.4224	0.11
2.00	0.8718	0.8797	0.03
3.00	1.3231	1.2459	0.01

Volatility dependence. The threshold is approximately constant ($\tau \approx 0.14$) across three orders of magnitude in $\sigma \in [0.05, 50]$. A power-law fit gives $\tau \propto \sigma^{0.016}$ with $R^2 = 0.43$ (poor fit), confirming Proposition 3.8.

Horizon dependence. The threshold is approximately constant ($\tau \approx 0.21$) for $h \in \{3, 10, 30, 100\}$. A power-law fit gives $\tau \propto h^{-0.009}$ with $R^2 = 0.13$ (poor fit), confirming Proposition 3.9.

Architecture dependence. Table 2 reports the exponents α, β (from fits $\tau \propto \sigma^{-\alpha} h^{-\beta}$) across five MDP architectures. All exponents are within 0.04 of zero, regardless of state-space size, temperature, or spectral gap of the transition matrix.

TABLE 2. Estimated exponents α and β across MDP architectures. All values are negligibly small.

Architecture	$\|\mathcal{S}\|$	T	α	β	Spectral gap
Tiny	24	0.3	0.023	0.039	0.088
Small	48	0.5	−0.018	−0.021	0.038
Medium	100	0.5	0.018	0.013	0.013
Cold	100	0.1	−0.021	0.004	0.013
Warm	100	1.0	0.027	−0.021	0.013

4. Discussion

4.1. **Refutation of the power-law hypothesis.** The original hypothesis that the misalignment threshold follows $\tau \propto \sigma^{-\alpha} h^{-\beta}$ with $\alpha, \beta > 0$ is refuted for softmax-parameterized trading agents. The numerically estimated exponents satisfy $|\alpha| < 0.03$ and $|\beta| < 0.04$ across all tested architectures, consistent with the theoretical prediction of Propositions 3.8 and 3.9 that both exponents vanish asymptotically. The correct characterization is the temperature scaling law of Corollary 3.6:

$$\tau(\delta) = T\sqrt{\frac{2\delta}{C(Q,\Phi)}} + O(\delta), \tag{4.1}$$

where $C(Q,\Phi) = \sum_s d(s)\ \mathrm{Var}_{\pi_0(\cdot|s)}(\Phi(s,\cdot))$ depends on the Q-function and perturbation direction but is independent of T at leading order.

The physical content of this result is as follows: the softmax function provides a built-in sensitivity scale governed by T. Perturbations to the reward propagate through the Q-values and into the policy via the softmax gradient, which scales as $1/T$. The Fisher information, being quadratic in the gradient, scales as $1/T^2$. Market parameters σ and h affect the Q-values, but since the softmax depends only on Q-differences (Remark 2.5), these effects are absorbed into the constant $C(Q,\Phi)$ without altering the dominant T-scaling.

4.2. **Connections to existing work.** *Impossibility of unhackability.* Skalse et al. [4] proved that over any open set of stochastic policies, every non-trivial proxy reward is hackable. Our Theorem 3.1 is consistent with this result: the upper bound $\tau < \infty$ means that misalignment *can* always be induced by a sufficiently strong perturbation. Our contribution is to quantify "sufficiently strong" via the Fisher information.

Distortion inevitability. Wang and Huang [5] showed that behavioral distortion is inevitable when the evaluation space has lower dimension than the true quality space. Their Proposition 1 establishes that the equilibrium effort profile e^* differs from the first-best e^{FB}. Our Theorem 3.4 adds a quantitative prediction: the magnitude of deviation grows as $\varepsilon^2/(2T^2)$ (from the leading term of the KL expansion), providing a formula absent from their framework.

Scaling laws for adversarial robustness. Debenedetti et al. [1] found that adversarial accuracy scales as $(\mathrm{FLOPs})^{0.01}$—a power law with a very small exponent. Our finding that the volatility exponent $\alpha \approx 0.016$ is similarly small suggests a common phenomenon: robustness metrics are insensitive to "scale" parameters (compute for neural networks, volatility for trading MDPs), while being primarily governed by "architecture" parameters (model design for networks, temperature for softmax policies).

Log-linear scaling of adversarial harm. Nathanson et al. [3] observed that mean harm scales logarithmically with the attacker-to-target size ratio. In our framework, the analogous quantity is the perturbation-to-temperature ratio ε/T: the KL divergence scales as $(\varepsilon/T)^2$, so the "harm" (measured by D) depends quadratically on this ratio rather than logarithmically. The discrepancy likely reflects the difference between finite MDP analysis and the complex dynamics of LLM-to-LLM interactions.

4.3. **Practical implications.** *Safety auditing.* Our results suggest that safety audits for LLM trading agents should prioritize measuring the effective policy

temperature—or more generally, the concentration of the decision distribution—rather than characterizing market conditions. The threshold formula $\tau = \sqrt{2\delta/\|G\|_F^2}$ provides a concrete, computable safety metric: given a desired significance level δ, one can compute the minimum perturbation intensity that would cause detectable misalignment.

Temperature-aware deployment. Corollary 3.6 implies that deploying agents with higher policy temperatures provides greater robustness to misalignment, at the cost of reduced decisiveness (Remark 3.7). In high-stakes trading environments, this suggests a principled approach to setting temperature: choose T large enough that $\tau(\delta) = T\sqrt{2\delta/C}$ exceeds the maximum plausible perturbation intensity, then accept the resulting entropy penalty.

Monitoring. Since τ depends primarily on T and the perturbation variance C, real-time monitoring of these quantities could provide early warning of misalignment risk. A sudden increase in C (due to a change in the effective perturbation landscape) would lower τ without any change in market conditions.

4.4. **Open questions.** We conclude with several directions for future work.

Conjecture 4.1 (Non-softmax parameterizations)**.** *For Gaussian policies $\pi(a \mid s) \propto \exp(-(a-\mu(s))^2/(2\sigma_\pi^2))$, the misalignment threshold satisfies $\tau \propto \sigma_\pi$ where σ_π is the policy standard deviation. More generally, for any exponential-family policy with natural parameter η, we conjecture $\tau \propto 1/\sqrt{I(\eta)}$ where $I(\eta)$ is the Fisher information of the family.*

(i) *Multi-agent composition.* In trading systems with multiple interacting LLM agents [6], how do individual thresholds τ_i compose into a system-level threshold? The Fisher information of a product of independent policies is additive, suggesting $\tau_{\text{system}}^{-2} = \sum_i \tau_i^{-2}$, but agent interactions may break this.

(ii) *Goodhart–Campbell transition.* Wang and Huang [5] conjecture a phase transition between gaming and degrading regimes. Our framework detects misalignment through KL divergence, which is a smooth function of ε. Can a discontinuity in $D(\varepsilon)$ or its derivatives signal such a transition?

(iii) *Dynamic perturbations.* We studied static perturbation directions Φ. For time-varying or adversarially adaptive perturbations, the threshold may decrease, since the adversary can exploit the agent's temporal correlations. A minimax formulation $\tau = \inf_\Phi \tau(\Phi)$ would capture the worst case.

(iv) *Finite-sample estimation.* In practice, D_{KL} is estimated from finite trajectory samples. The estimation error introduces a "noise floor" below which misalignment is undetectable. Characterizing the interaction between estimation error and the threshold τ is an important practical question.

(v) *Tighter convergence rates.* The rates $O(\sigma^{-2})$ and $O(h^{-1})$ in Propositions 3.8 and 3.9 may not be tight. In particular, for very small σ (highly concentrated transitions), the behavior of $\tau(\sigma)$ may be qualitatively different.

5. Conclusion

We developed a mathematical framework for misalignment sensitivity thresholds in softmax-parameterized trading agents. The central object is the threshold $\tau(\delta)$, the minimum perturbation intensity producing KL divergence exceeding δ. Our

main theorem characterizes this threshold via the Fisher information of the policy parameterization: $\tau = \sqrt{2\delta/\|G\|_F^2} + O(\delta)$.

The principal finding is the temperature scaling law $\tau \propto T$ (Corollary 3.6), which identifies the softmax temperature as the dominant determinant of misalignment robustness. This result, confirmed computationally with $R^2 = 0.999$, has a clear practical interpretation: agents with more concentrated decision distributions (lower T) are more vulnerable to reward perturbations. Market volatility and decision horizon are asymptotically irrelevant (Propositions 3.8 and 3.9), refuting the initially hypothesized power-law dependence on these parameters.

These results redirect attention from external market conditions to internal policy architecture. The threshold formula provides a concrete, computable safety metric for deployment decisions: given a maximum plausible perturbation intensity and a desired significance level, one can determine the minimum temperature required for robust operation. Future work should extend the framework to non-softmax parameterizations, multi-agent systems, and dynamic perturbation models.

References

1. Edoardo Debenedetti, Zishen Wan, Maksym Andriushchenko, et al., *Scaling compute is not all you need for adversarial robustness*, arXiv preprint arXiv:2312.13131 (2023).
2. Zixuan Liu and Thanh Dang, *FINRS: A risk-sensitive trading framework*, arXiv preprint arXiv:2511.12599 (2025).
3. Samuel Nathanson, Cynthia Matuszek, and Rebecca Williams, *Scaling patterns in adversarial alignment*, arXiv preprint arXiv:2511.13788 (2026).
4. Joar Skalse, Nikolaus Howe, Dmitrii Krasheninnikov, and David Krueger, *Defining and characterizing reward hacking*, Advances in Neural Information Processing Systems, 2022.
5. Jiacheng Wang and Jinbin Huang, *Reward hacking as equilibrium under finite evaluation*, arXiv preprint arXiv:2603.28063 (2026).
6. Chenghua Xiao et al., *TradingAgents: Multi-agents LLM financial trading framework*, arXiv preprint arXiv:2412.20138 (2024).